

Basis & Dimension, Rank & Nullity

Null space

✓ **Basis:** If V is a vector space & $S = \{v_1, v_2, \dots, v_n\}$ is a set of vectors in V , then S is called a basis for V if the following 2 conditions hold:

- * a) S is linearly independent (one solution) ^{H.E.} (linear combination of vectors)
- * b) S spans V

Example Proof that the vectors: $(1, 2, 0)$, $(0, 5, 7)$ & $(-1, 1, 3)$ form a basis of $\mathbb{R}^3$.

Solution The given vectors are basis iff they are linearly independent & $\mathbb{R}^3$ & span.

① L.I.

$$x(1, 2, 0) + y(0, 5, 7) + z(-1, 1, 3) = (0, 0, 0)$$

$$\begin{bmatrix} 1 & 0 & -1 & | & 0 \\ 2 & 5 & 1 & | & 0 \\ 0 & 7 & 3 & | & 0 \end{bmatrix} \quad x=0, y=0, z=0$$

So, the vectors are L.I.

② $(a, b, c) = x(1, 2, 0) + y(0, 5, 7) + z(-1, 1, 3)$

$$\begin{bmatrix} 1 & 0 & -1 \\ 2 & 5 & 1 \\ 0 & 7 & 3 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \end{bmatrix} = \begin{bmatrix} 1 & 0 & -1 & | & a \\ 0 & 1 & 3/5 & | & (b-2a)/5 \\ 0 & 0 & 1 & | & (7b-5c-14a)/6 \end{bmatrix}$$

which implies the vectors span $\mathbb{R}^3$
 $\therefore$ These vectors form the basis for $\mathbb{R}^3$.

** Basis — set of vectors
 # of vectors in a basis = ~~dimension~~ rank
 independent

①

Row Space & Column Space

If A is an $m \times n$ matrix, then the subspace of $\mathbb{R}^n$ spanned by the row vectors of A is called the row space of A , & the subspace of $\mathbb{R}^n$ spanned by the column vectors is called the column space of A .

The solution space of the homogeneous system of eqns $Ax=0$, which is a subspace of $\mathbb{R}^n$, is called the nullspace of A .

Ex Find the basis for the row & column space of

$$A = \begin{bmatrix} 2 & 2 & 0 & 4 & 0 & 4 \\ 0 & 2 & 4 & 4 & 0 & 2 \\ 1 & 1 & 6 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ -1 & 1 & -2 & 3 & 0 & 2 \end{bmatrix}$$

$$\begin{aligned} R_1' &= R_1 - R_3 \\ R_2' &= \frac{R_2}{2} \\ R_3' &= R_4 - 2R_3 \\ R_5' &= R_5 + R_3 \end{aligned}$$

$$\begin{bmatrix} 1 & 1 & -6 & 3 & 0 & 4 \\ 0 & 1 & 2 & 2 & 0 & 1 \\ 0 & 0 & -12 & 2 & 0 & 4 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 2 & 4 & 4 & 0 & 2 \end{bmatrix}$$

row echelon form
row operation
after 3 steps

$$\begin{bmatrix} 1 & 1 & -6 & 3 & 0 & 4 \\ 0 & 1 & 2 & 2 & 0 & 1 \\ 0 & 0 & 1 & -\frac{1}{6} & 0 & -\frac{1}{3} \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Thm If a matrix R is in row echelon form, then the row vectors with the leading 1's forms the basis for the row space of R & the column vectors with leading 1's of the row vectors form a basis for the column space of R .

The non-zero row vectors of R form the basis for the row space R , and hence form the row space of A . The basis vectors are

$$\begin{aligned} r_1 &= (1, 1, 6, 1, 0, 0) \\ r_2 &= (0, 1, 2, 2, 0, 1) \\ r_3 &= (0, 0, 1, -\frac{1}{6}, 0, -\frac{1}{3}) \end{aligned}$$

(2)

The vectors

$$c'_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}, c'_2 = \begin{bmatrix} 1 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}, c'_3 = \begin{bmatrix} 6 \\ 2 \\ 1 \\ 0 \\ 0 \end{bmatrix} \left. \vphantom{\begin{matrix} c'_1 \\ c'_2 \\ c'_3 \end{matrix}} \right\} \text{ form a basis for the col. space of } R.$$

~~form a basis for the column space of R,~~
thus the corresponding column vectors of A, namely

$$c_1 = \begin{bmatrix} 2 \\ 0 \\ 1 \\ 0 \\ -1 \end{bmatrix}, c_2 = \begin{bmatrix} 2 \\ 2 \\ 2 \\ 0 \\ 1 \end{bmatrix}, c_3 = \begin{bmatrix} 0 \\ 4 \\ 6 \\ 0 \\ -2 \end{bmatrix} \left. \vphantom{\begin{matrix} c_1 \\ c_2 \\ c_3 \end{matrix}} \right\} \text{ form a basis for the col. space of A.}$$

~~form a basis for the column space of A.~~

of independent vectors in a basis

Rank The common dimension of the row & col. space of matrix A is called the rank of A.

The rank of A = $\dim(\text{row space / col. space}) = 3$

~~by the one 3 in this case they are different.~~

$$\text{Rank}(A) = \dim(\text{row space } A) = \dim(\text{col. space } A)$$

Dimension Given a basis of a vector space, every element of the vector space can be expressed uniquely as a finite linear combination of basis, and all bases of a vector space have the same # of elements, called the dimension of the vector space.

OR The dimension of a v.s. is equal to the maximum no. of linearly independent vectors contained in it.)

(3)

Solution Space / Null space

" " The solution space of the homogeneous system of eqns $Ax=0$, which is a ~~subspace~~ of $\mathbb{R}^n$, is called the nullspace of A .

Ex find the basis of the solution space of the matrix

$$A = \begin{bmatrix} -1 & 2 & 0 & 4 & 5 & -3 \\ 3 & -7 & 2 & 0 & 1 & 4 \\ 2 & -5 & 2 & 4 & 6 & 1 \\ 4 & -9 & 2 & -4 & -4 & 7 \end{bmatrix}$$

" " $\downarrow$
reduced
row
echelon form
 $\text{rank}(A)=2$

$$\begin{bmatrix} 1 & 0 & -4 & -28 & -37 & 13 \\ 0 & 1 & -2 & -12 & -16 & 5 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\underline{Ax = 0 \text{ Form}}$$

$$\Rightarrow x_1 - 4x_3 - 28x_4 - 37x_5 + 13x_6 = 0$$

$$x_2 - 2x_3 - 12x_4 - 16x_5 + 5x_6 = 0$$

2 ~~equations~~ equations & 6 variables

" " 4 free variables. Let x_3, x_4, x_5, x_6 are free variables.

$$x_1 = 4x_3 + 28x_4 + 37x_5 - 13x_6$$

$$x_2 = 2x_3 - 12x_4 + 16x_5 - 5x_6$$

let $x_3 = r, x_4 = s, x_5 = t, x_6 = 4$

$x_1 = 4r + 28s + 37t - 13u, x_4 = s$

$x_2 = 2r + 12s + 16t - 5u, x_5 = t$

$x_3 = r, x_6 = 4$

or equivalently,

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \\ x_6 \end{bmatrix} = r \begin{bmatrix} 4 \\ 2 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} + s \begin{bmatrix} 28 \\ 12 \\ 0 \\ 1 \\ 0 \\ 0 \end{bmatrix} + t \begin{bmatrix} 37 \\ 16 \\ 0 \\ 0 \\ 1 \\ 0 \end{bmatrix} + u \begin{bmatrix} -13 \\ -5 \\ 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}$$

The four vectors form a basis of the solution space.

Nullity: The dimension of the solution space of A is called the nullity of A and denoted by nullity(A).

rank(A) = 4 in previous example.

Dimension Theorem If A is a matrix of n columns, then, rank(A) + nullity(A) = n

rank(A) = 2

nullity(A) = 4

$n = 6$ } There are 6 columns in matrix A

The null space has dimension 4

* The column space of a matrix is called the range of A

* The dimension of the nullspace of A is called the nullity of A